

Energy Performance Prediction for the Energy Transition in France

Machine Learning Models for Building Energy Assessment
and Renovation Decision Support



ADEME Dataset



50k+ Observations

Presentation Overview

1 Context and strategic importance

2 Challenges

3 Our Solution Framework

4 Data Overview and Methodology

5 Model and Business Impact

6 Limitations and Next Steps

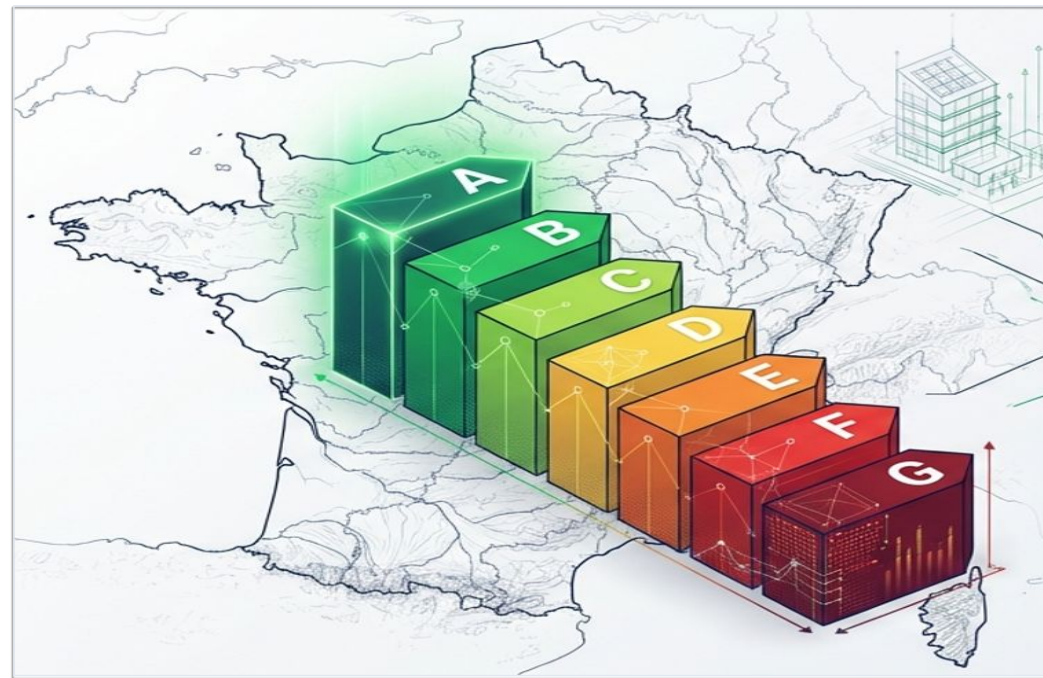
Energy Performance Assessments: A Key Challenge for the Green Transition



Context

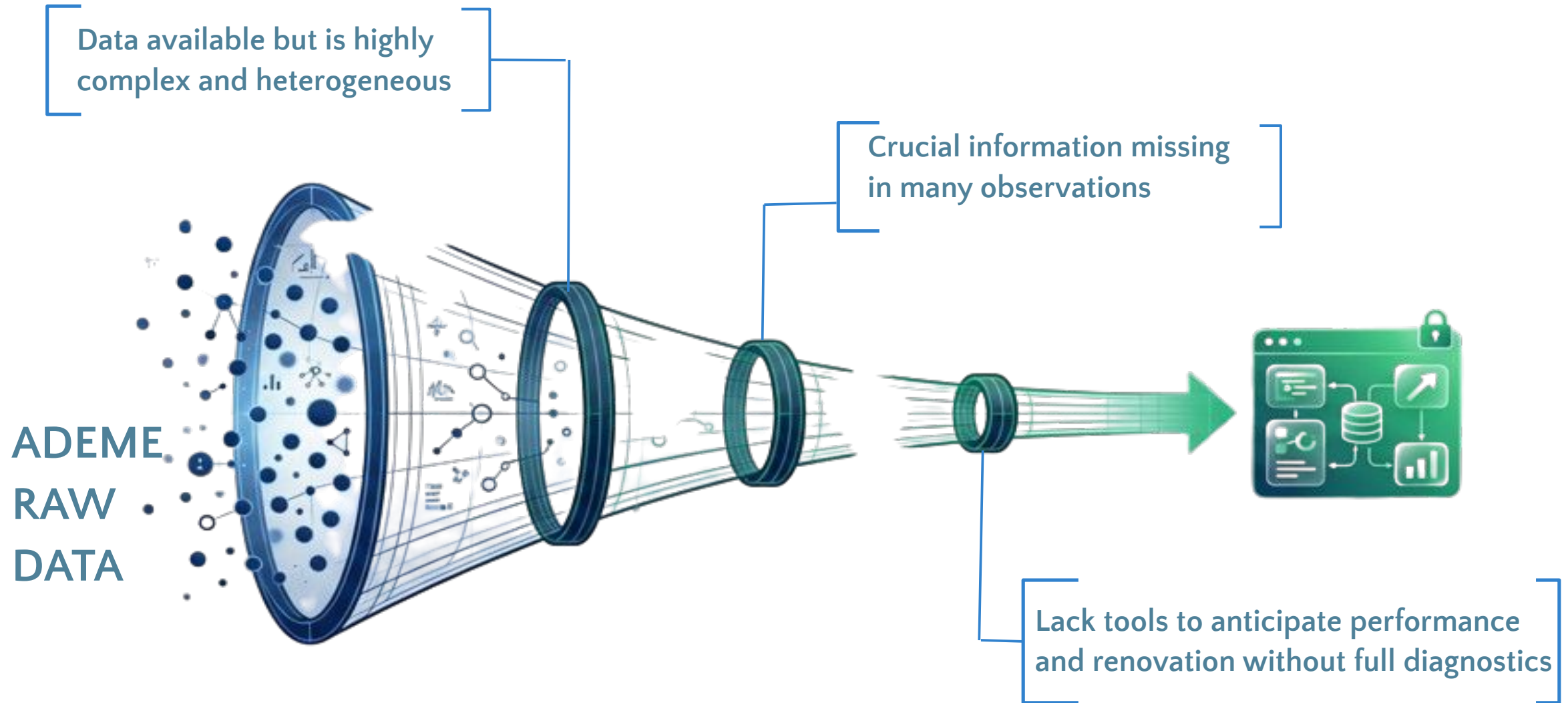
Energy Performance Certificates (DPE) and (GHG) emissions data are critical regulatory tools for assessing building energy consumption across France.

- ✓ Essential for real estate transactions
- ✓ Required for renovation grants
- ✓ Monitors energy efficiency policies



Since 2020 ADEME has published an open database containing millions of energy performance assessments covering all buildings types across the entire French territory, a real opportunity to transform these observations into an active predictive level

Challenges



Complete predictive Ecosystem for renovation decisions



Predict Energy Efficiency (EPC label)

Multi-class Classification

Predicts the energy consumption, helps identifying the most energy-intensive buildings.

🎯 Target: DPE label (A to G)



Predict GHG Emissions

Multi-class Classification

Estimates greenhouse gas emissions, targeting renovations with highest environmental impact.

🎯 Target: GHG label (A to G)



Recommend Renovation Actions

Multi-class Classification

Guide renovation decisions by predicting the most relevant improvement categories for each building.

🎯 Target: Recommendation categories



Estimate Post Renovation Cost

Regression

Assess financial benefits and help decision-makers plan investments with accurate cost projections.

🎯 Target: Costs after renovation

Data Overview

An introduction to the dataset powering our housing energy analysis across France.

Dataset Characteristics

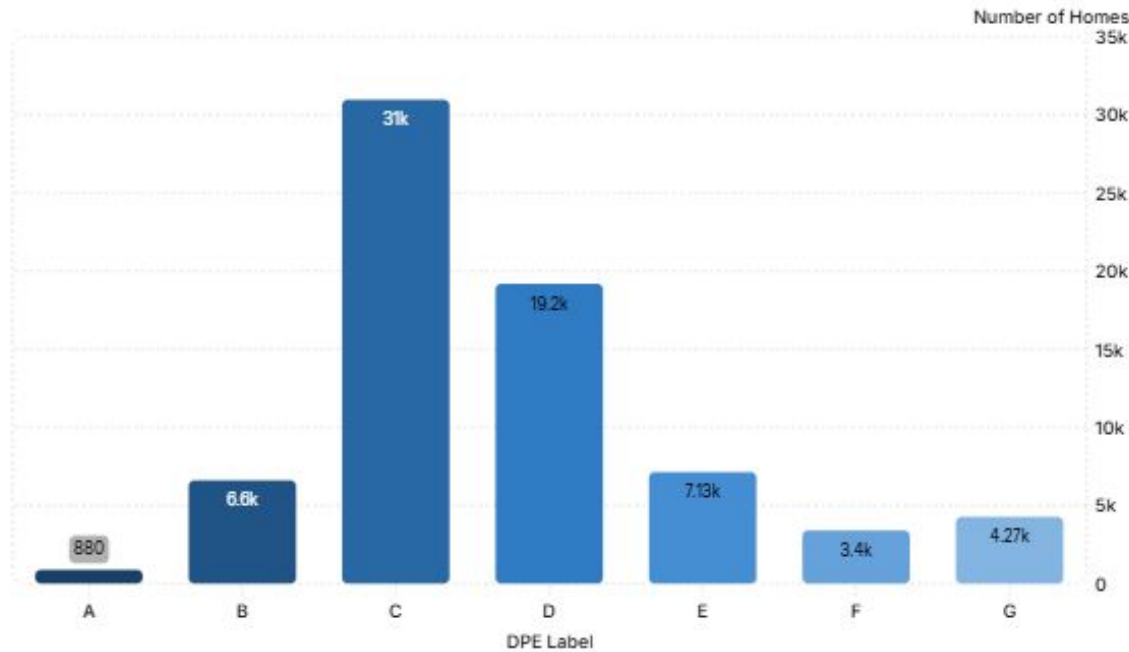
Source ADEME Open Data	Size 50,889 housing records
Coverage Across all of France	Building Types All types of dwellings

Key Variables

Living Area	Energy Consumption
Heating Demand	Heating Type
Climate Zone	Construction Year

Key Insights from the Data

Before building any models, we explored the data to understand what drives energy performance in French housing.



Classes C and D are the most common. F and A the least efficient homes are far less represented.

3 Key Findings

1

Heating is the biggest driver

Homes with higher heating demand consistently score lower on energy efficiency.

2

Location shapes energy needs

Climate zone plays a major role, colder regions require significantly more energy to heat homes.

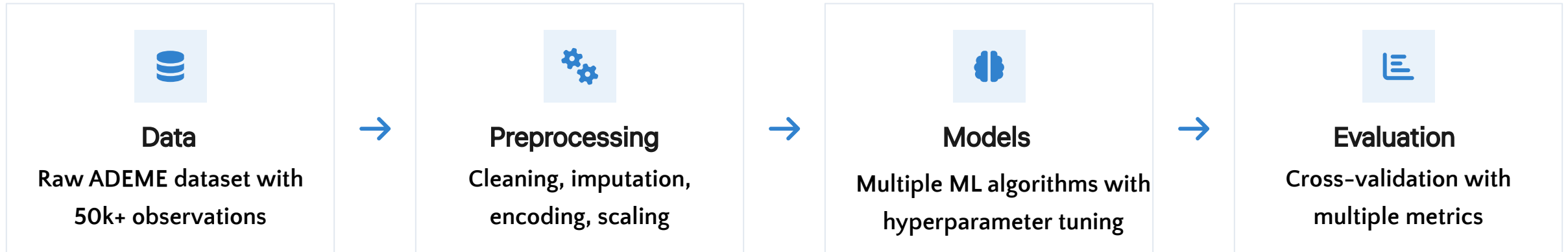
3

Data is not evenly distributed

Some energy classes are much more common than others, which adds complexity to making accurate predictions.

Methodology

End-to-end machine learning pipeline



Preprocessing

Missing Data Handling

Advanced imputation methods based on missing data type (MCAR, MAR, MNAR):

- ✓ KNN Imputation
- ✓ Multiple Regression (Ridge)
- ✓ MICE (Iterative Imputer)

Feature Engineering

Feature transformation applied :

- ✓ Normalization & Standardization
- ✓ One-Hot & Ordinal Encoding
- ✓ Feature Selection (RF, VIF)

Class Imbalance

We handled imbalanced target distributions:

- ✓ SMOTE (Synthetic Oversampling)
- ✓ Class Weight Balancing

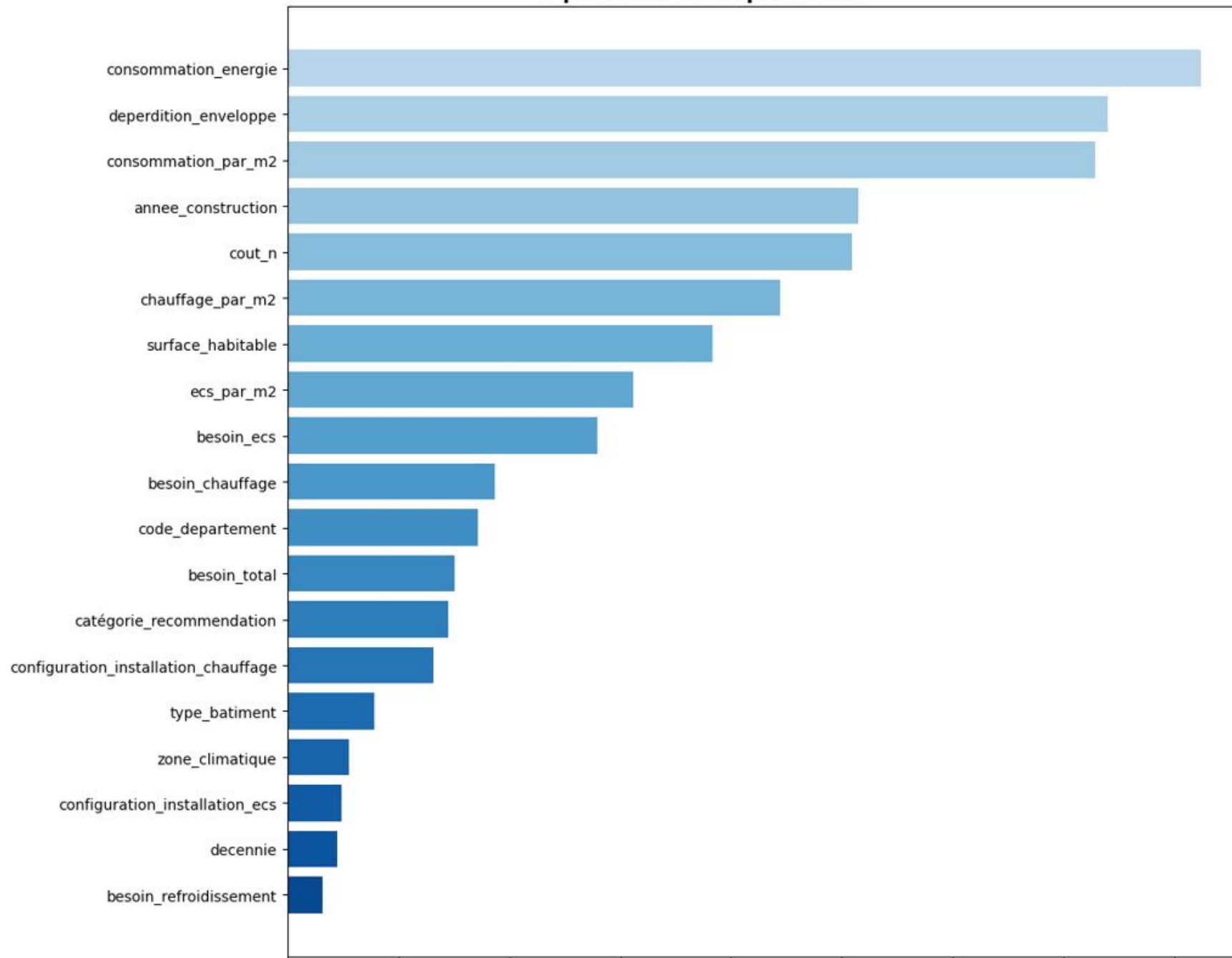


Split Train/Val/Test: 70% / 15% / 15%



Cross-Validation: 5-Fold Stratified

Top 30 Feature Importances



Predicting the DPE Label

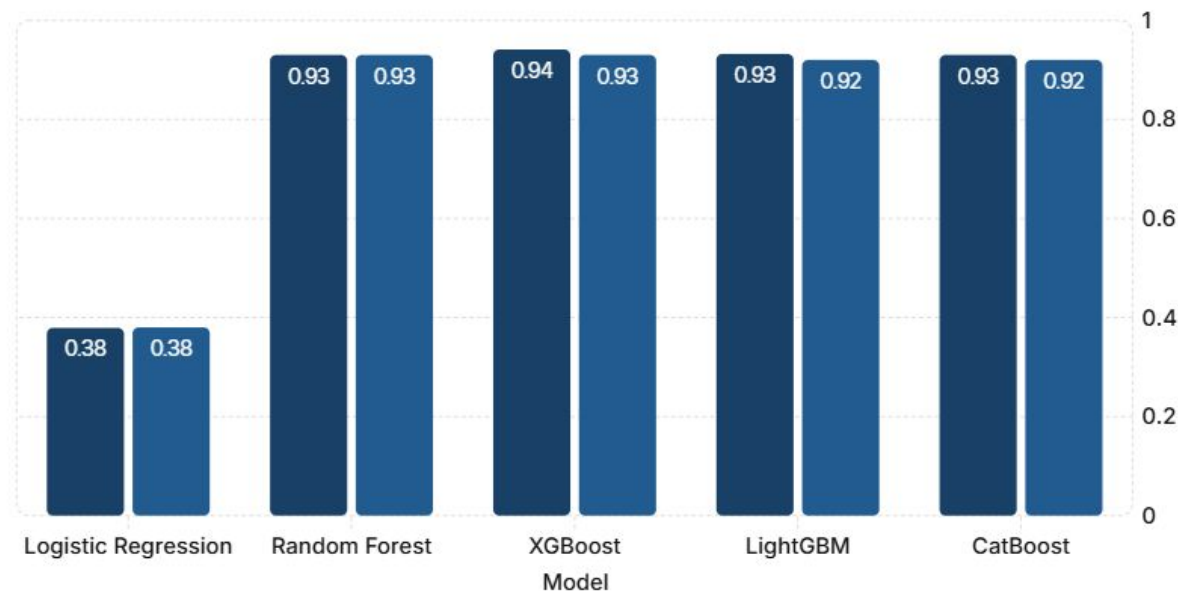
Goal : predict the DPE label of each home.

We tested several models with many combinations of hyperparameters to see which one gave the best results.

The four models have very close results, but XGBoost does a little better. It reached an accuracy of 94%. This means the model predicts the correct DPE label in about 94 cases out of 100.

Its Macro F1 score is 0.93, which shows that the model performs well not only on the most common categories, but across all DPE classes.

■ Test Accuracy ■ Macro F1



This prediction can bring several practical benefits:

- Save time because a full DPE assessment can take a long time
- Reduce costs because a full assessment can be expensive
- Prioritize renovations by identifying the homes that need action first

Best Model: XGBoost

Accuracy

94.08%

Best overall performance

Best results among all tested models

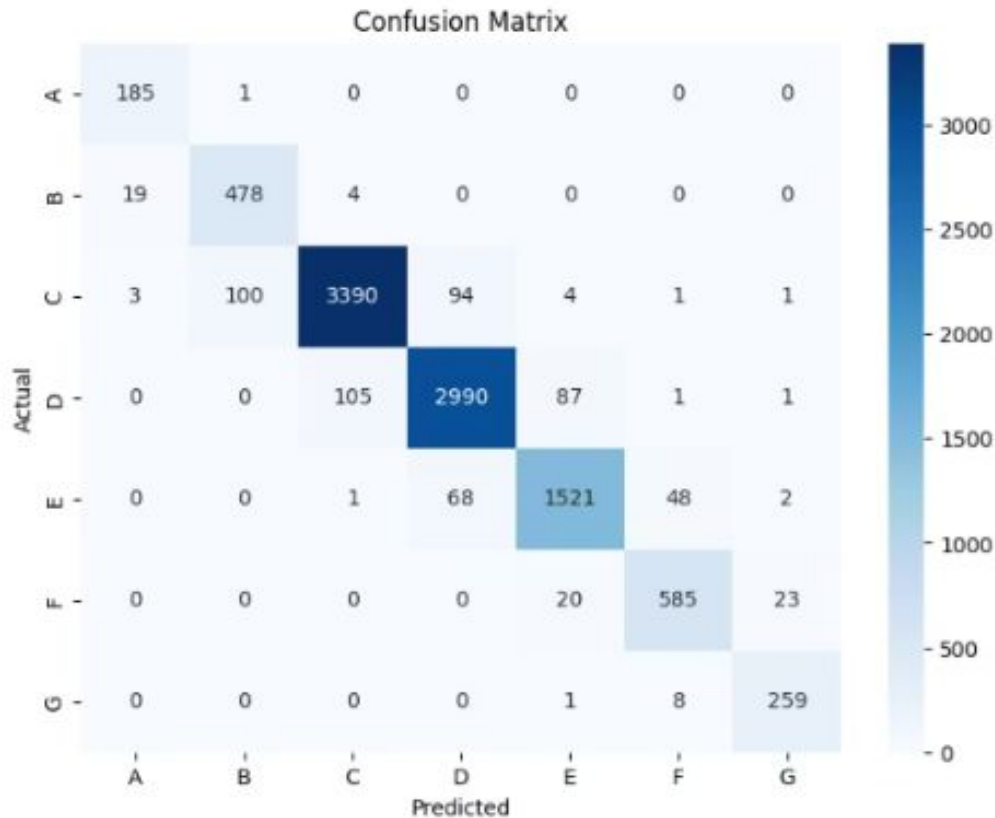
Reliable decision-support tool

Useful for practical decision-making

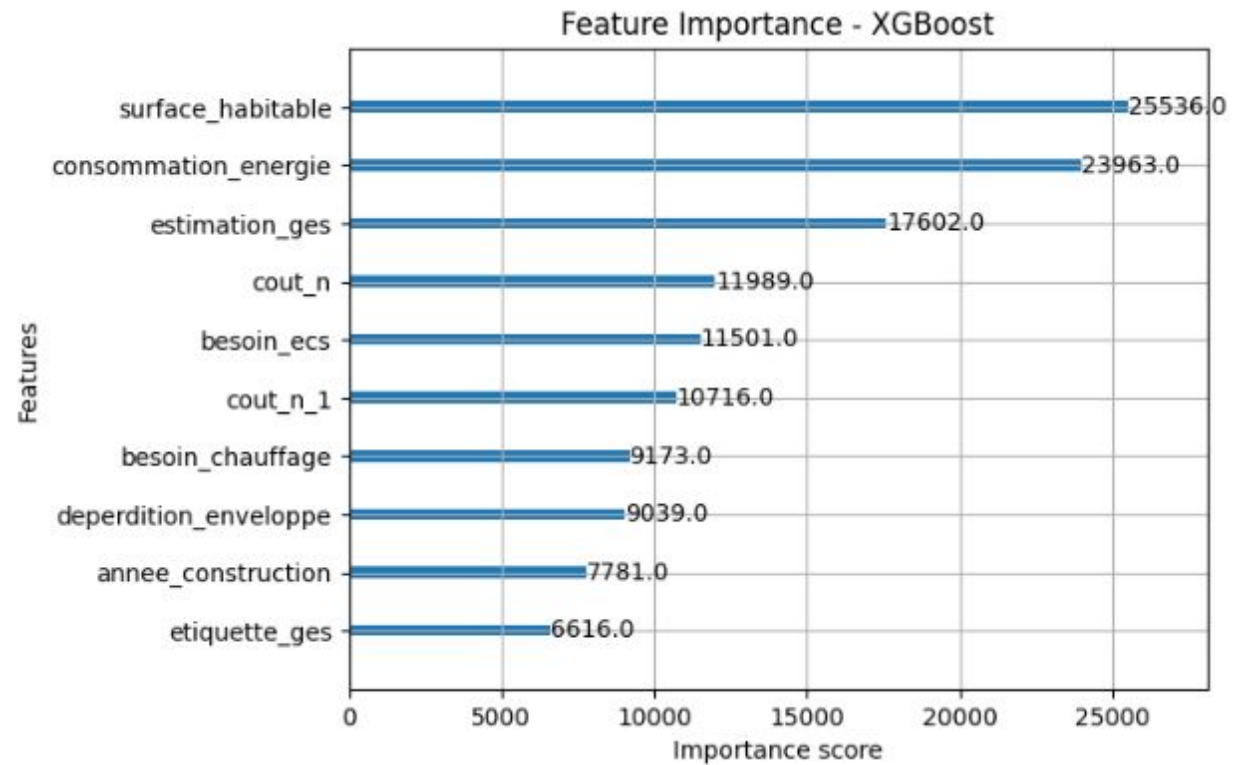
We also tested models like SVM and MLP, but the dataset was too large, so the training could not be completed.

What the Results Show

Confusion Matrix



The confusion matrix shows that most predictions are correct. When the model makes mistakes, they usually happen between similar labels such as D and C or D and E and E and F.



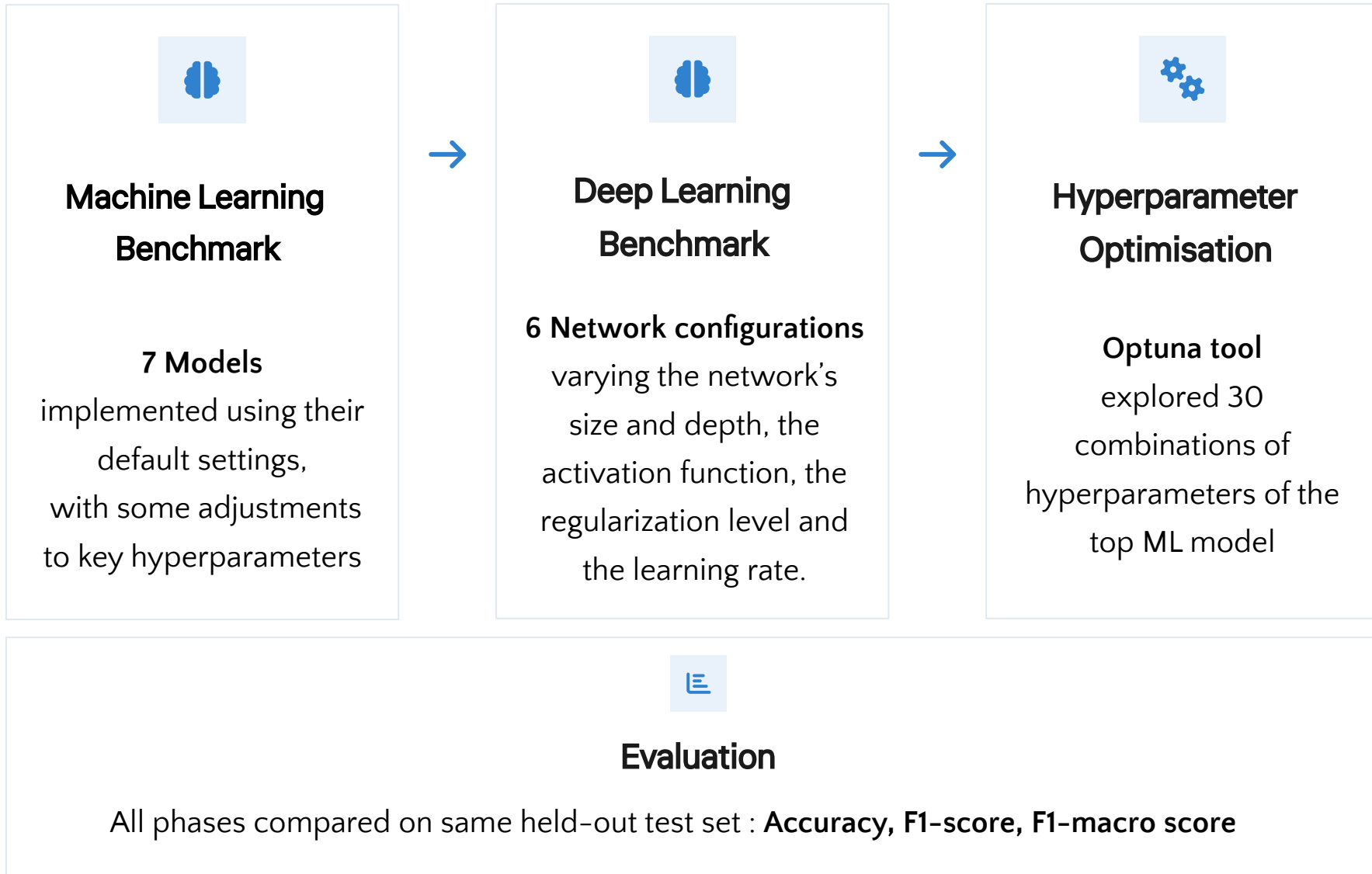
We also looked at the main factors behind the prediction. The most important ones are living area, energy consumption, and GHG estimate.

Key Takeaway

- 📌 The model is highly reliable and performs well in practice.

GHG Emissions Prediction

Methodology and experimental configurations



Experimental configurations

ML Model	Key settings
LightGBM	objective='multiclass', num_class=7, random_state=42
CatBoost	iterations=500, random_state=42, verbose=0
Random Forest	n_estimators=200, random_state=42
Extra Trees	n_estimators=200, random_state=42
Gradient Boosting	n_estimators=200, random_state=42
HistGradientBoosting	random_state=42

DL Configuration	Architecture	Activation	Alpha	Batch size	Learning rate	Max iter	Early stopping
Config 1 – Small	(64, 32)	ReLU	0.0001	128	0.0013	300	Yes
Config 2 – Medium	(128, 64, 32)	ReLU	0.0001	256	0.0014	0	Yes
Config 3 – Large	(256, 128, 64, 32)	ReLU	0.0001	256	0.0015	0	Yes
Config 4 – Strong Reg	(128, 64, 32)	ReLU	0.001	256	0.0014	0	Yes
Config 5 – Low LR	(128, 64, 32)	ReLU	0.0001	128	0.00055	0	Yes
Config 6 – Tanh	(128, 64, 32)	Tanh	0.0001	256	0.0014	0	Yes

Results and analysis ML banchmark

ML Model	CV Acc Mean	CV Acc Std	Train Acc	Test Acc	Test F1
HistGradientBoosting	0.8202	0.0018	0.8598	0.8238	0.8243
Gradient Boosting	0.8014	0.0031	0.8362	0.8056	0.8061
LightGBM	0.7994	0.0027	0.8205	0.8008	0.8012
CatBoost	0.7831	0.0023	0.7955	0.7844	0.7848
Random Forest	0.7667	0.0024	0.8353	0.7694	0.7696
Extra Trees	0.6341	0.0026	0.6797	0.6311	0.6263
Logistic Regression	0.4465	0.0061	0.4410	0.4408	0.4239

Results and analysis DL banchmark

DL Configuration	Test Accuracy	Test F1-Score	Overfitting
Config 3 – Large Network	0.8334	0.8338	0.0359
Config 4 – Strong Regularization	0.8303	0.8305	0.0227
Config 5 – Low Learning Rate	0.8272	0.8277	0.0280
Config 2 – Medium Network	0.8261	0.8266	0.0221
Config 6 – Tanh Activation	0.8248	0.8251	0.0194
Config 1 – Small Network	0.8143	0.8146	0.0099

Results and analysis Tuning best ML model

Tuning hyperparameter of HistGradientBoosting

```
max_iter = 550, max_depth = 25, learning_rate ≈  
0,064, min_samples_leaf = 50, l2_regularization  
= 2,6, max_bins = 255, max_leaf_nodes = 127 et  
n_iter_no_change = 15.
```



Best ML Model: HistGradientBoosting

Test Accuracy

84.49%

Macro F1

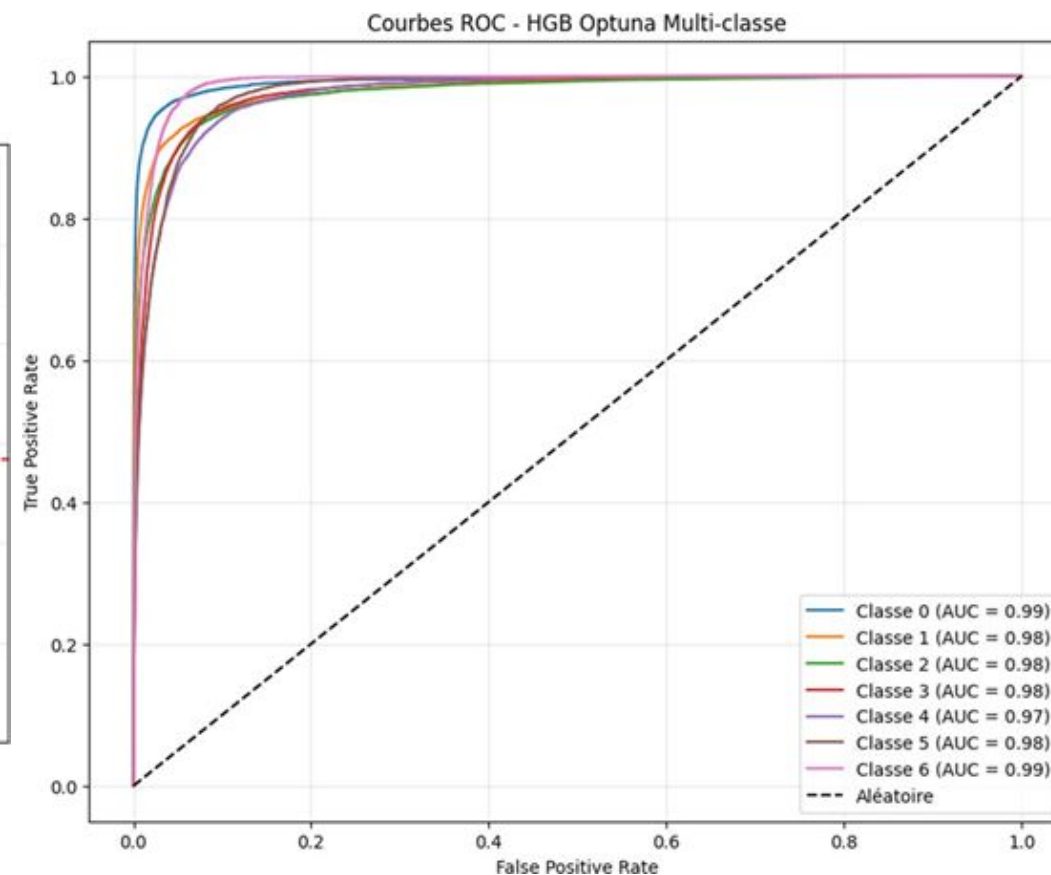
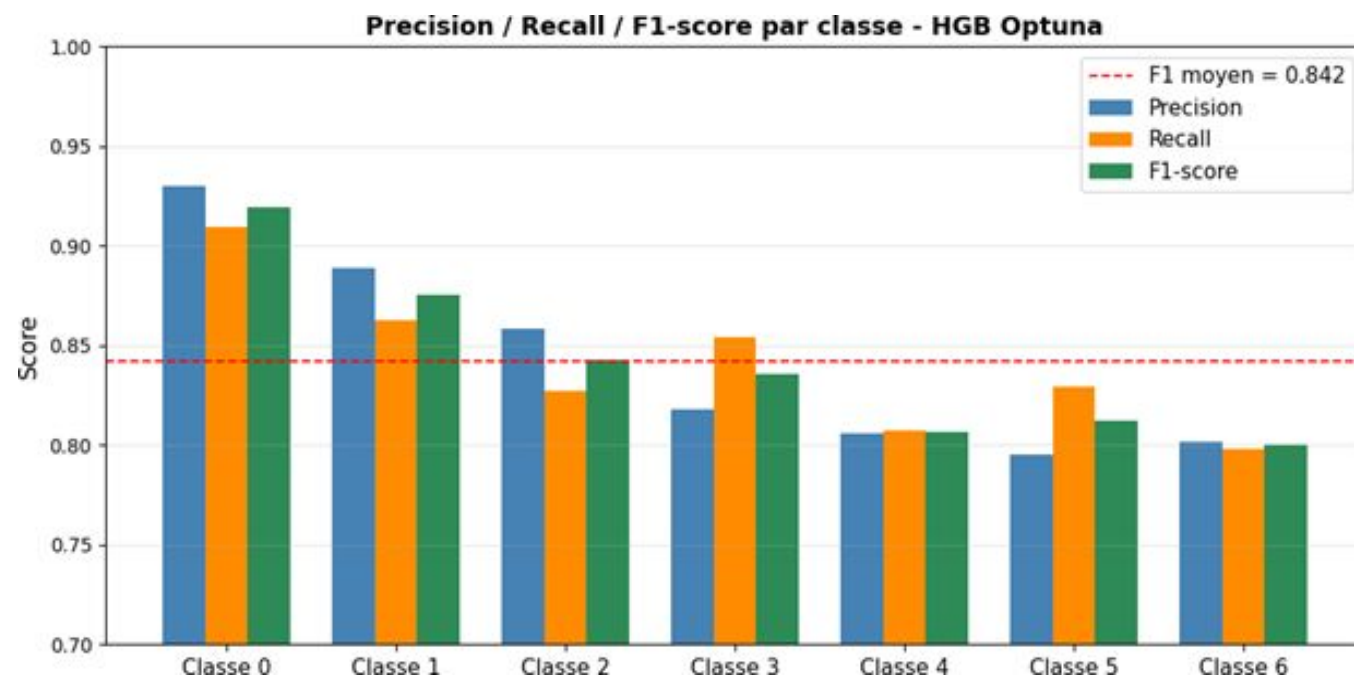
0.842

Macro AUC

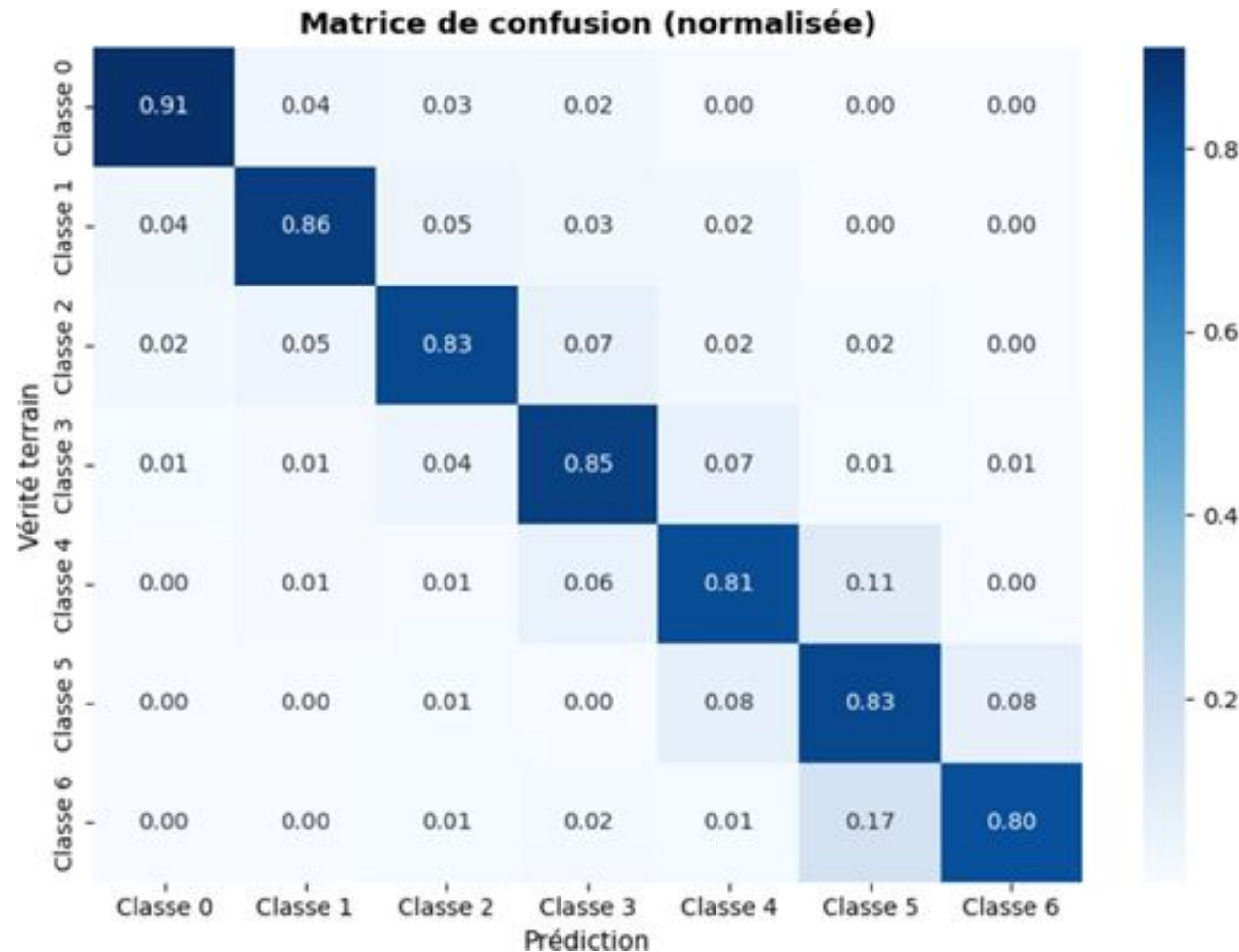
0.9819

Micro AUC

0.9831



Results and analysis



The models developed can predict residential greenhouse gas (GHG) emission classes with a high degree of accuracy ($\approx 84\%$), making them a valuable tool for decision-making.

A detailed examination of performance by category shows excellent reliability for Class A, which has the lowest error rate (9.2%)

The other classes have error rates ranging from 13.4% (Class D) to 18.2% (Class E), confirming overall robustness across the entire emissions spectrum

Post-Renovation Energy Cost Estimation

Need 3 · Predict annual post-renovation energy cost (€/year)

Objective

Predict how much energy will cost after renovation — turning energy performance into a clear financial estimate for homeowners, investors, and public stakeholders.

MAE

~103 €

Average error per prediction

R²

0.98

Variance explained by the model

Model comparison — test set performance

Model	MAE (€)	RMSE (€)	R ²
✓ Deep Neural Network (selected)	103	—	0.98 ★
Random Forest	74	571	0.98
XGBoost	157	831	0.96

Why the Neural Network? Although Random Forest has lower raw MAE (74 €), the neural network shows near-identical stability across training, validation and test sets (MAE ≈ 105 → 103), indicating better generalization and lower overfitting risk.

In practice: instead of only saying this home will be more efficient — we can now say: here is the expected impact on the energy bill.

Model Training Stability Analysis

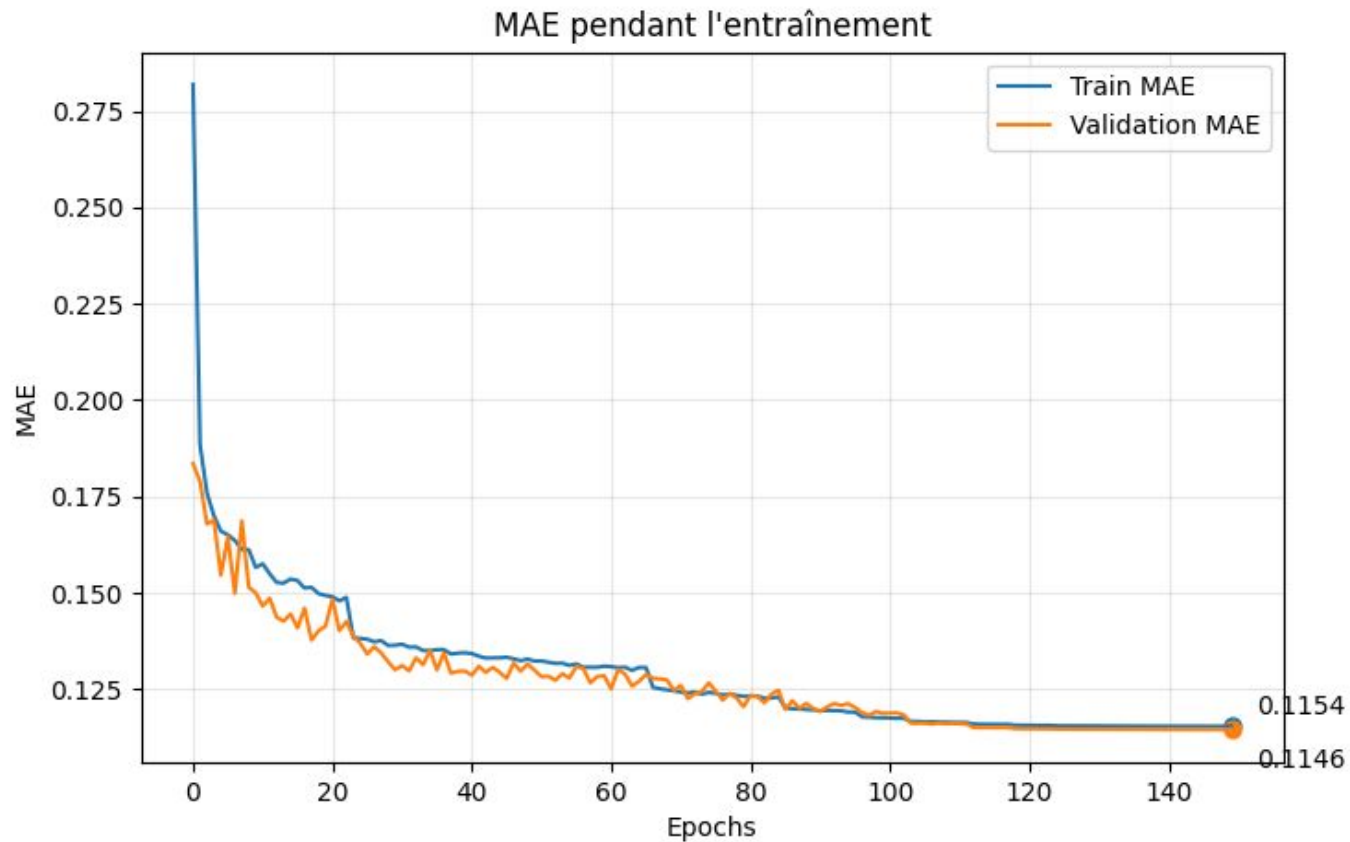


Figure 1: Mean Absolute Error (MAE) during model training — Training vs Validation curves

- **Model Stability**

This graph shows that training and validation remain very close, so the model is stable.

- **Error Interpretation**

The value around **0.11** is the error on the scaled target.

€ Original Unit Conversion

Once converted back to the original unit, it corresponds to about **103 euros**.

Limitations & Critical Analysis

Need 3 · What to keep in mind when using the model

01

Data quality drives prediction quality

- Missing values were imputed using regression and MICE techniques
- 26% of `cout_n` values were reconstructed — introducing some uncertainty
- When input data is incomplete, predictions carry a margin of error

02

Performance vs. robustness trade-off

- Random Forest achieves lower raw MAE (74 €) than the neural network (103 €)
- However it shows less consistent behaviour across train / val / test splits
- We chose the neural network for its stability — not just its raw score

Key takeaway: we selected models based on overall stability, not just one metric — this ensures more reliable real-world predictions.

How we built the recommendation model

Need 4 — Renovation recommendation prediction

Our approach

- Predict the most relevant renovation action for each dwelling (Heating , insulation, ventilation etc.)
- Tested several machine learning models
- Primary metric: Macro F1, a balanced score because some categories are rare

Models tested

★ Random Forest (selected)
XGBoost
CatBoost
Neural Network

Model comparison — Accuracy & Macro F1 - We're keeping the best

Model	Valid. Acc.	Test Acc.	Macro F1 (test)
✓ Random Forest (baseline)	0.4339	0.4320	0.3028 ★
XGBoost	0.4347	0.4327	0.3008
Neural Network	0.4268	0.4251	0.2558
CatBoost	0.2827	0.2847	0.2616

Best balance
between
performance and
stability



What the Model Really Does Well (and Not)

Strong Performance

The model performs **well** on the most frequent renovation categories:

Heating	F1: 0.48
Walls / Insulation	F1: 0.49
Ventilation	F1: 0.44

Struggles with Rare Categories

The model **struggles with rare categories**, especially specific recommendations:

Plumbing / Sanitary	F1: 0.01
Floors	F1: 0.17
Technical Improvements	F1: 0.19

Why This Happens

1 Data distribution

Some renovation types are naturally more common than others.

→ Most buildings receive general recommendations (heating, insulation).

2 Rare recommendations

More specific recommendations are less frequently assigned in real cases.

→ They appear much less in the dataset.

3 The model reflects real-world data patterns.

→ It performs better on frequent renovation cases, which are more represented in the dataset.

Understanding model limitations

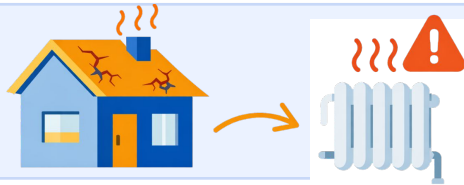
Confusion patterns & dataset bias

Why errors happen

The model confuses similar renovation categories :

Heating ↔ Insulation

→ *Poor insulation leads to heating problems*



Domestic hot water (ECS) ↔ Heating

→ The same system is often used for both heating and hot water

Roof / Ceilings ↔ Insulation

→ *Roof is part of the insulation envelope*

These errors are not random, they reflect real-world situations where a dwelling has multiple renovation needs at once.



Important: Dataset Limitation

In the original data:

A dwelling can have multiple renovation recommendations

In our dataset:

Only the main recommendation is kept per dwelling

This creates a bias:

The model can be **penalized even when its prediction is actually relevant**, if it's the second or third recommendation in the original data.

Business Impact of our models

Identify Energy-Inefficient Homes

→ Use EPC and GHG predictions to identify the most energy-consuming buildings

- ✓ 94% accuracy in EPC classification
- ✓ 84% accuracy in GHG prediction
- ✓ Prioritize Class G & F buildings

Estimate Costs & Financial Impact

- Estimate energy costs and potential savings
- Help users and investors plan budgets and investments

Recommend Renovation Actions

- Suggest the most relevant renovation actions
- Mainly for the most common building situations

Our models help identify problems and estimate potential savings

Next Steps for Improvement

How to improve this approach

01

Improve dataset quality

- Include all recommendations per dwelling
- Better reflect real building situations

02

Handle data imbalance

- Collect more data for rare cases
- Improve predictions for minority categories

03

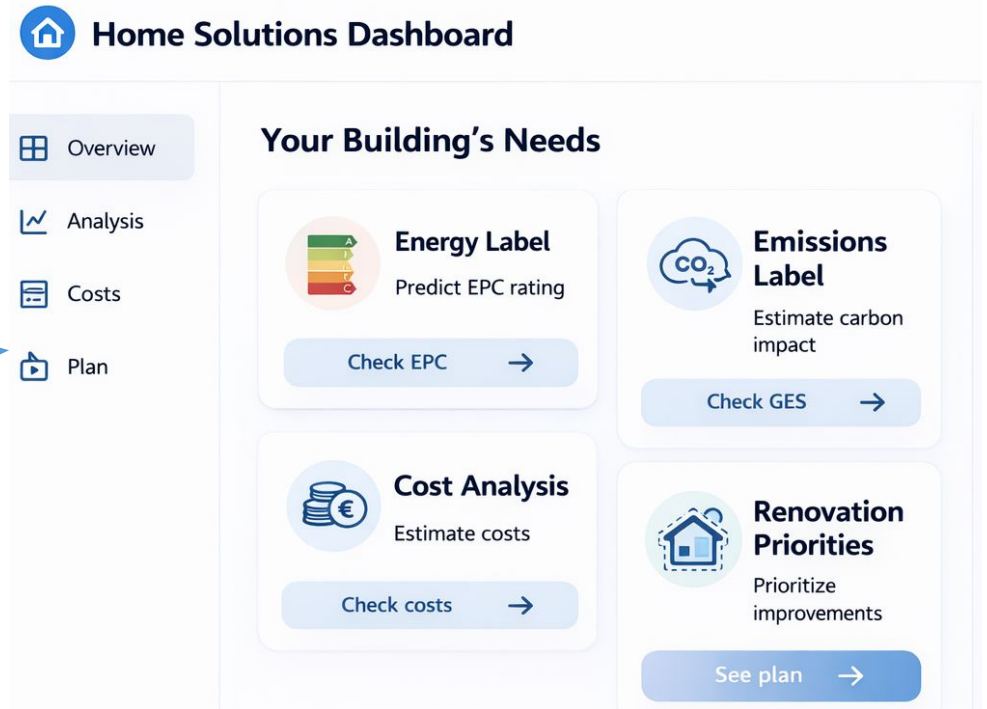
Create a simple interface to

- Estimate energy performance, emissions, costs, and renovation needs
- Support decision-making for users and professionals

04

Integrate real-time energy data

- to keep the models updated and improve predictions over time.



Conclusion

We developed predictive models to better understand building energy performance

Our approach allows to:

- **Predict energy ratings and emissions**
- **Estimate energy costs**
- **Identify renovation needs**

In practice, this helps:

- **Identify inefficient buildings**
- **Prioritize renovation actions**
- **Support better decision-making**

**Thank you for
your attention**

